

1. INTRODUCTION

The RoboInject-300 is a fully electric injection molding machine (serial prefix SM-) manufactured by RoboMech Industries. It features a servo-driven injection arm with a peak clamping force of 300 tons and four independent barrel heating zones. This service manual covers all units with control system version 4.3 and above.

1.1 SAFETY SYMBOLS

WARNING ? Risk of electrical shock or personal injury. Failure to follow instructions may result in death or severe injury.

CAUTION ? Risk of damage to equipment or minor injury.

NOTE ? Additional information to aid in operation.

1.2 TECHNICAL SPECIFICATIONS

Power supply: 480V 3-phase, 60A

Servo motor rating: 15 kW continuous, 22 kW peak

Barrel zones: 4 × 1500W heater bands

Max barrel temperature: 350°C

Cooling: forced air via 230V AC fan (0.25 kW)

Controller: RoboMech RM-430 with 7-inch HMI

Weight: 4,200 kg

1.3 LUBRICATION SYSTEM

The central lubrication system uses ISO-VG-68 grease. The reservoir holds 1.5 L and the pump delivers 0.5 mL per cycle to the ball screw and linear guides. Low oil level triggers alarm A107 on the HMI display. Refill port is located behind the front access panel, secured with two M6 hex bolts.

1.4 MAINTENANCE SCHEDULE

Daily: inspect cooling fan intake for debris, check oil level, verify no alarm codes on the HMI.

Weekly: clean air filter, verify zone temperatures within $\pm 5^{\circ}\text{C}$ of setpoint, check servo drive cooling fan operation.

Monthly: inspect injection arm ball screw for wear, check belt tension, perform sensor calibration per §6.0.

Quarterly: replace air filter, inspect all electrical connections, test safety interlocks per §7.3.

Annually: full calibration, replace heating bands if resistance deviates more than 10% from nominal, replace lubrication pump filter.

6. SENSOR CALIBRATION PROCEDURES

6.1 NTC Winding Temperature Sensor

The winding temperature sensor is a negative-temperature-coefficient (NTC) thermistor. Calibration should be performed annually or whenever the sensor is replaced. Use the following reference table:

Temperature	Resistance (nominal)	Tolerance
25°C	2.40 k?	±5%
50°C	1.10 k?	±8%
75°C	0.55 k?	±8%
100°C	0.28 k?	±10%
120°C	0.18 k?	±10%

To calibrate: immerse the sensor tip in a calibrated oil bath at the reference temperature and compare the measured resistance. If the reading deviates by more than the tolerance, replace the sensor (P/N NTC-R300). Ensure the sensor tip is fully inserted into the thermal well and coated with heat-conductive paste (P/N TCP-100).

6.2 Type-K Thermocouple Calibration (Barrel Zones)

Each barrel zone uses a Type-K thermocouple with a measurement range of 0-400°C. Cold junction compensation is handled automatically by the controller board. If zone temperature readings are erratic or deviate by more than ±10°C from setpoint:

Step 1. Swap the suspect thermocouple with an adjacent zone to isolate whether the fault follows the sensor or the controller channel.

Step 2. If the fault follows the sensor, replace the thermocouple (P/N TC-K-300).

Step 3. If the fault stays on the same channel, the controller input board may need replacement.

6.3 Pressure Transducer Calibration

The injection pressure transducer (range 0-250 bar, output 4-20 mA) should be zeroed annually. With the system at atmospheric pressure, adjust the 'Zero' potentiometer on the transducer signal conditioner to read 4.00 mA at the controller input. The full-scale reading should be 20.00 mA at 250 bar.

7. ELECTRICAL SYSTEM

7.1 MAIN FUSE RATINGS

Main AC input (L1, L2, L3): 3 × 32A class gG, P/N F-M300-MAIN

Servo drive input: 2 × 16A class aM, P/N F-M300-SERVO

Control circuit transformer primary: 1 × 6.3A class gG, P/N F-M300-CTRL

Heater zone contactors: 1 × 25A per zone class gG

24V DC power supply: 1 × 4A class gG

WARNING: Always disconnect and verify zero energy state before accessing the fuse panel. The DC bus capacitors can retain lethal charge for up to 5 minutes after power-down.

7.2 CONTROL BOARD STATUS LEDS

D1 (green, steady) ? 24V DC control power present

D2 (yellow, flashing) ? CAN bus communication active

D3 (red, steady) ? Fault latch active (see error code on HMI)

D4 (amber, flashing) ? Maintenance reminder (service interval due)

7.3 SAFETY INTERLOCK CIRCUIT

All access doors are monitored by magnetic safety switches (MSS-300, 2NC contacts each). The interlock circuit is dual-channel, monitored by the safety PLC. Both channels must close within 500 ms of each other for the machine to start. If the controller shows 'E011 ? Safety Loop Open' on the HMI diagnostics screen:

Step 1. Open and close each access door one at a time while watching the I/O status on the HMI (Diagnostics > I/O Map > Safety Inputs).

Step 2. Identify the door switch that does not toggle.

Step 3. Replace the faulty switch (P/N MSS-300). Torque the mounting screws to 3 N·m. Align the actuator with the center of the switch body.

7.4 WIRING DIAGRAM REFERENCE

Full wiring diagrams are provided in Appendix A, sheets A-1 through A-12. Key connections:

TB1-1,2: 480V AC main input

TB2-1,2: 24V DC fan supply

TB3-1 through 4: heater zone outputs

J3-1 through 4: NTC sensor input

J5-1 through 8: safety interlock inputs